

terminal, *t*
metavarroot, *mvr*
nontermroot, *ntr*
suffix, *suff*
auxfn, *f*
prodname, *pn*
index, *i*, *j*, *n*, *m*

<i>metavar</i> , <i>mv</i>	::=	<i>metavarroot</i> <i>suffix</i>
<i>nonterm</i> , <i>nt</i>	::=	<i>nontermroot</i> <i>suffix</i>
<i>element</i> , <i>e</i>	::=	<i>terminal</i> <i>metavar</i> <i>nonterm</i>
<i>mse</i>	::=	<i>metavar</i> <i>auxfn</i> (<i>nonterm</i>) <i>mse</i> union <i>mse'</i> {}
<i>bindspec</i> , <i>bs</i>	::=	bind <i>mse</i> in <i>nonterm</i> <i>auxfn</i> = <i>mse</i>
<i>prod</i> , <i>p</i>	::=	<i>element</i> ₁ .. <i>element</i> _{<i>m</i>} :::: <i>prodname</i> (+ <i>bindspec</i> ₁ .. <i>bindspec</i> _{<i>n</i>} +)
<i>rule</i> , <i>r</i>	::=	<i>nontermroot</i> :: ' ' :: = <i>prod</i> ₁ .. <i>prod</i> _{<i>m</i>}
<i>grammar_rules</i> , <i>g</i>	::=	grammar <i>rule</i> ₁ .. <i>rule</i> _{<i>m</i>}
<i>auxfn_type</i> , <i>at</i>	::=	<i>nontermroot</i> ₁ .. <i>nontermroot</i> _{<i>n</i>} → <i>metavarroot</i>
<i>auxfn_type_env</i> , <i>Φ</i>	::=	empty <i>Φ</i> , <i>auxfn</i> : <i>auxfn_type</i>
<i>grammar_type</i> , <i>Γ</i>	::=	<i>nontermroot</i> ₁ .. <i>nontermroot</i> _{<i>n</i>}
<i>formula</i>	::=	<i>judgement</i> <i>i</i> = <i>j</i> <i>f</i> = <i>f'</i> <i>mvr</i> = <i>mvr'</i> <i>ntr</i> = <i>ntr'</i> <i>mv</i> = <i>mv'</i> <i>nt</i> = <i>nt'</i> <i>e</i> = <i>e'</i>

		$mse = mse'$ $bs = bs'$ $p = p'$ $r = r'$ $g = g'$ $(formula)$ $\mathbf{not\ formula}$ $\forall i \in 1..m. formula$ $\exists i \in 1..m. formula$ $formula \wedge formula'$ $formula \Rightarrow formula'$ $\mathbf{true}$
<i>terminals</i>	$::=$	$\{\}$ $(+$ $+))$ $, ,$ $::$ $::=$ $\mathbf{grammar}$ $\mathbf{bind}$ $\mathbf{in}$ $\mathbf{union}$ $\rightarrow$ $\Rightarrow$ $\vdash$ $\wedge$ $\vee$ $ $
<i>Jtype</i>	$::=$	$\Phi \vdash f : at$ $\Phi; e_1 .. e_n \vdash mse : metavarroot$ $\Phi; e_1 .. e_n \vdash bs \mathbf{ok}$ $\Phi \vdash prod \mathbf{ok}$ $\Phi \vdash rule \mathbf{ok}$ $\Phi \vdash grammar_rules \mathbf{ok}$
<i>judgement</i>	$::=$	$Jtype$
<i>user_syntax</i>	$::=$	$terminal$ $metavarroot$ $nontermroot$ $suffix$ $auxfn$ $prodname$ $index$

$\mid$ *metavar*
 $\mid$ *nonterm*
 $\mid$ *element*
 $\mid$ *mse*
 $\mid$ *bindspec*
 $\mid$ *prod*
 $\mid$ *rule*
 $\mid$ *grammar_rules*
 $\mid$ *auxfn_type*
 $\mid$ *auxfn_type_env*
 $\mid$ *grammar_type*
 $\mid$ *formula*
 $\mid$ *terminals*

$\Phi \vdash f : at$

$$\frac{}{\Phi, f : at \vdash f : at} \text{FF_1}$$

$$\frac{\Phi \vdash f : at \quad \mathbf{not}(f = f')}{\Phi, f' : at' \vdash f : at} \text{FF_2}$$

$\Phi; e_1 .. e_n \vdash mse : metavarroot$

$$\frac{}{\Phi; e_1 .. e_n \vdash \{\} : mvr} \text{MSE_1}$$

$$\frac{e_j = mvr \text{ suff}}{\Phi; e_1 .. e_n \vdash mvr \text{ suff} : mvr} \text{MSE_2}$$

$$\frac{\Phi \vdash f : ntr_1 .. ntr_m \rightarrow mvr \quad e_j = nt \quad nt = ntr_i \text{ suff}}{\Phi; e_1 .. e_n \vdash f(nt) : mvr} \text{MSE_3}$$

$$\frac{\Phi; e_1 .. e_n \vdash mse : mvr \quad \Phi; e_1 .. e_n \vdash mse' : mvr}{\Phi; e_1 .. e_n \vdash mse \mathbf{union} mse' : mvr} \text{MSE_4}$$

$\Phi; e_1 .. e_n \vdash bs \mathbf{ok}$

$$\frac{\Phi; e_1 .. e_n \vdash mse : mvr \quad e_j = nt}{\Phi; e_1 .. e_n \vdash \mathbf{bind} mse \mathbf{in} nt \mathbf{ok}} \text{Bs_1}$$

$$\frac{\Phi; e_1 .. e_n \vdash mse : mvr \quad \Phi \vdash f : ntr_1 .. ntr_n \rightarrow mvr}{\Phi; e_1 .. e_n \vdash f = mse \mathbf{ok}} \text{Bs_2}$$

$\Phi \vdash prod \mathbf{ok}$

$$\frac{\forall i \in 1..m. \Phi; e_1 .. e_n \vdash bs_i \mathbf{ok} \quad prod = | e_1 .. e_n :::: prodname(+bs_1 .. bs_m+)}{\Phi \vdash prod \mathbf{ok}} \text{PROD_1}$$

$\Phi \vdash rule \mathbf{ok}$

$$\frac{\forall i \in 1..m. \Phi \vdash \text{prod}_i \mathbf{ok} \quad \text{rule} = \text{ntr}::'':::=\text{prod}_1 .. \text{prod}_m}{\Phi \vdash \text{rule} \mathbf{ok}} \quad \text{RULE_1}$$

$$\boxed{\Phi \vdash \text{grammar_rules} \mathbf{ok}}$$

$$\frac{\begin{array}{l} \forall i \in 1..m. \Phi \vdash \text{rule}_i \mathbf{ok} \\ \text{grammar_rules} = \mathbf{grammar} \text{ rule}_1 .. \text{rule}_m \\ \forall i \in 1..m. \forall j \in 1..n. ((\text{rule}_i = \text{ntr}::'':::=\text{prod}_1 .. \text{prod}_m \wedge \text{rule}_j = \text{ntr}::'':::=\text{prod}'_1 .. \text{prod}'_n) \Rightarrow i = j) \end{array}}{\Phi \vdash \text{grammar_rules} \mathbf{ok}} \quad \text{GRAMM}$$

Definition rules: 11 good 0 bad

Definition rule clauses: 30 good 0 bad